

SingleRAN

RTOS Security Feature Parameter Description

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Security Declaration

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For vulnerability information, enterprise customers can visit the following web page:

<https://securitybulletin.huawei.com/enterprise/en/security-advisory>

Contents

1 Change History.....	1
1.1 SRAN20.1 01 (2024-03-09).....	1
1.2 SRAN20.1 Draft A (2023-12-31).....	1
2 About This Document.....	2
2.1 General Statements.....	2
2.2 Applicable RAT.....	2
3 Overview.....	4
3.1 Overview.....	4
3.2 Security Threats and Security Requirements.....	4
3.3 Security Architecture.....	5
4 Security Features.....	7
4.1 System User Management.....	7
4.2 File System and Permission Management.....	7
4.2.1 File and Directory Access Control.....	7
4.2.2 File Protection.....	8
4.3 Network Management.....	8
4.3.1 Protocols Enabled by Default.....	8
4.3.2 Services Enabled by Default.....	8
4.4 Malware Prevention Policies.....	9
4.5 OS Integrity Protection.....	9
4.5.1 Software Release Integrity Protection.....	9
4.5.2 Software Loading Integrity Protection.....	9
4.6 System and Security Log Management.....	10
4.6.1 Log Files.....	10
4.6.2 Log Auditing.....	10
4.7 System Upgrade and Patch Policy.....	10
4.8 Other Security Hardening Policies.....	11
4.8.1 Support for SELinux Configurations.....	11
4.8.2 Address Space Layout Randomization.....	11
4.8.3 System Commissioning.....	11
5 Glossary.....	12

6 Reference Documents.....13

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN20.1 01 (2024-03-09)

This issue does not include any changes.

1.2 SRAN20.1 Draft A (2023-12-31)

This issue introduces the following changes to SRAN17.1 02 (2022-04-27).

Technical Changes

Change Description	Parameter Change	Base Station Model
Added support for the BSC6960. For details, see related descriptions in the document.	None	None

Editorial Changes

Changed *Base Station RTOS Security* to *RTOS Security*.
Revised descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements of the operating environment that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to base stations and the BSC6960.

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and NR.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

3 Overview

3.1 Overview

The real-time operating system (RTOS) is a Linux-based operating system tailored to provide full security protection for telecommunications products. As part of an end-to-end security architecture and OS-level security hardening solution, the RTOS is enhanced in hardware support, commissioning, and performance optimization to minimize security risks. This design and optimization provides all-round security experience for products.

The customized RTOS consists of the kernel and root file system:

- Kernel: The kernel is customized and includes the latest patch, which helps improve system security.
- Root file system: The RTOS is a compact OS. The root file system only contains necessary component files and service files to minimize security risks.

3.2 Security Threats and Security Requirements

The main threats on RTOS security are security vulnerabilities, unauthorized operations, and information disclosure. [Table 3-1](#) describes these threats.

Table 3-1 Main security threats to RTOS

Threat	Description	Severity	Security Requirement
Security vulnerabilities	The kernel and service processes have known security vulnerabilities.	Major	Fixed versions of software, including the full OS patch, are released according to the promised repair plan. The software is upgraded to fix the system security vulnerabilities.

Threat	Description	Severity	Security Requirement
Unauthorized operations	Unauthorized users log in to the RTOS and obtain unauthorized rights.	Major	• User login is not allowed. • Security-Enhanced Linux (SELinux) and system and resource access authorization configurations are supported. • File and directory security configuration • Log tracing and auditing
Information disclosure	Insecure services	Major	Unnecessary services are disabled. Only necessary and secure services are provided for software.

 NOTE

The BSC6960 does not support SELinux.

3.3 Security Architecture

RTOS functions as a bridge between hardware resources (multi-core CPUs and other hardware devices) and services. As a multi-process OS running on mid-range and high-end multi-core CPUs, RTOS features a security architecture incorporating the security policies listed in [Table 3-2](#).

Table 3-2 Security architecture of RTOS

System User Management	• System user management
File System and Permission Management	• File and directory access control • File protection
Network Management	• Protocols enabled by default • Services enabled by default
Malware Prevention Policies	• Malware prevention policies
OS Integrity Protection	• Software release integrity protection • Software loading integrity protection
System and Security Log Management	• Log file • Security auditing

System Upgrade and Patch Policy	• System upgrade and patch policy
Other Security Hardening Policies	• Support for SELinux configurations • Address space layout randomization • System commissioning

 NOTE

The BSC6960 does not support SELinux.

4 Security Features

4.1 System User Management

The RTOS supports multiple users. Service processes are executed by common users, and system administration processes are executed by service users. Remote management is not allowed for any users. Least privilege needs to be configured based on function requirements.

- Common users: can execute service processes and cannot log in to the OS. They can create, modify, or delete files under their specific home directories. (For example, user **jack** can perform relevant operations under the home directory **/home/jack**.) In addition, common users can run scripts or binary executable files under the **/usr/bin** and **/bin** directories.
- Service users: can execute system administration processes and cannot log in to the OS. Service users are granted extremely low operation permission. This prevents unauthorized users from exploiting the vulnerabilities of system administration processes to destroy the system, reducing security risks.

4.2 File System and Permission Management

File system permission is categorized into read, write, and execute permission. Common users can operate only the files on which permission is granted. Permission management ensures file security.

4.2.1 File and Directory Access Control

File and directory permission specifies who can access and perform operations on these files and directories.

- Access permission on files and directories is categorized into: read-only, write-only, and executable.
- There are three types of users who can access these files and directories:
 - File owner: creator of the file by default
 - Group user: users in the same group as the file owner

- Other user: users in a different group from the file owner
- File access permission is set as required based on the least privilege principle. For example, if a non-executable file managed by a service process needs to be modified, the file permission is set to 640, indicated by the binary number 110100000. The meaning of the binary number is as follows:
 - The left-most 110 indicates that the file owner can read and write but cannot execute this file.
 - The middle 100 indicates that group users can read but cannot write or execute the file.
 - The right-most 000 indicates that other users cannot read, write, or execute the file.

4.2.2 File Protection

The RTOS restricts common users' access to system files.

- Common users cannot access the home directories of other users.
- Common users cannot modify or delete system commands and library files, device files (/dev), and configuration files (/etc).

NOTE

- The read permission on a directory indicates that a user can view the files and sub-directories under the directory. The write permission indicates that a user can create files and sub-directories under the directory. The execute permission indicates that a user can go to the directory.
- The read permission on a file indicates that a user can view the content in the file. The write permission indicates that a user can edit the content in the file. The execute permission indicates that a user can execute the commands in the file.

4.3 Network Management

4.3.1 Protocols Enabled by Default

By default, the User Datagram Protocol (UDP), Transmission Control Protocol (TCP), and Internet Control Message Protocol (ICMP) are enabled in the RTOS. The preceding protocols are used for internal communications only and are not used for that with other NEs.

4.3.2 Services Enabled by Default

[Table 4-1](#) lists the default services provided by the system.

Table 4-1 Default services provided by the system

Service Name	Available or Not	Description
syslog-ng/ rsyslog	Yes	This service is started in the inittab file and is used for log recording.

Service Name	Available or Not	Description
cron	Yes	The service is a daemon process for executing scheduling commands.
auditd	Yes	This service is used to save the audit information generated by the kernel to a log file (for example, <code>/var/log/audit/audit.log</code>).

4.4 Malware Prevention Policies

Malware prevention measures have been fully considered and enhanced for the software and internally encapsulated OS. No antivirus software needs to be deployed after delivery.

- The OS is an embedded OS. Unnecessary services have been disabled. Only necessary and secure services are provided for software.
- The OS does not support remote user management.
- The OS supports SELinux and has strict permission control over file systems.

NOTE

The BSC6960 does not support SELinux.

- Secure boot, software verification, and process auditing are supported to prevent malware from being implanted.

4.5 OS Integrity Protection

4.5.1 Software Release Integrity Protection

The RTOS components include the vmlinux (kernel) and initrd (root file system). The software architecture separates the kernel mode from the user mode to enhance system security.

The RTOS is encapsulated in software. Before the OS components and software are released, acknowledged antivirus software such as McAfee, Avira, OSCE, and Kav is used to scan the released components and delete viruses. This ensures that the released components are not infected with viruses.

4.5.2 Software Loading Integrity Protection

The RTOS is encapsulated in software. When the software is released, digital signature is performed. When the boards are powered on or reset, the digital signature must be verified before software loading. Illegal software cannot pass the digital signature verification and the software loading fails.

4.6 System and Security Log Management

Logs record system running information and are of vital importance to system security. Major log functions include auditing and monitoring. With logs, you can diagnose problems, monitor real-time system status, and track traces left by attackers.

4.6.1 Log Files

The following describes the common log files of the RTOS:

- messages: record kernel and system information by default.
- warn: records all warnings and error information of the system.
- wtmp: records all local and remote logins, changes in system running level, and time of the changes. This file is encrypted.
- auth.log: contains system authorization information, including user logins and permission mechanism.
- debug: records all debugging log files of the system.
- error: records all error information of the system.
- kern.log: contains logs generated by the kernel, which helps resolve problems during kernel customization.
- syslog: records the log files recorded by syslog-ng.
- user.log: records information about users of all levels.
- lastlog: records the latest login information of all users.

NOTE

The log files and OS are stored in different partitions. In addition, the dumping, scrolling, and polling mechanisms are used to prevent the log storage partitions from being used up.

4.6.2 Log Auditing

The RTOS does not provide direct access interfaces for external systems. Users can use the maintenance and test function provided by the base station software to obtain logs, locate faults, and audit security.

- Run the **ULD FILE** command with **SRCF** set to **BRDLOG(Compositive Log)** to upload logs (including OS log files).
- Perform security inspection.

4.7 System Upgrade and Patch Policy

For system security vulnerabilities, fixed versions of software, including the full OS patch, are released according to the promised repair plan. The software is upgraded to fix the system security vulnerabilities. Independent OS upgrades are not supported.

4.8 Other Security Hardening Policies

4.8.1 Support for SELinux Configurations

SELinux is short for Security-Enhanced Linux, which originated from the high guaranteed OS security and microkernel research since the 1980s. In December 2000, the United States National Security Agency (NSA) released its first public version. SELinux is a mandatory access control function. Based on the principle of least privilege, it uses Linux security modules in the OS kernel, tags all objects in the system, and configures access policies for these objects to implement access control.

The base station RTOS normalizes various resources (including processes) into files. It tags all resources and sets access policies based on the process function requirements and security isolation requirements to minimize the permission.

NOTE

The BSC6960 does not support SELinux.

4.8.2 Address Space Layout Randomization

Address space layout randomization (ASLR) is a security protection technology for buffer overflow. It randomizes the layout of the heap, stack, and shared library mapping (mmap and vdso (X86)) and increases the difficulty of attackers in predicting the destination address, preventing attackers from directly locating codes to guard against buffer overflow attacks. ASLR supports the ARM and X86 architectures.

Function dynamic switch: `/proc/sys/kernel/randomize_va_space`

- **0:** indicates that the function is disabled.
- **1:** indicates that this function is enabled, and the addresses of the stack and shared library mapping are randomized.
- **Other values:** indicate that this function is enabled, and the addresses of the heap, stack, and shared library mapping are randomized.

The value **2** is configured for the RTOS by default.

4.8.3 System Commissioning

The RTOS does not provide the commissioning function.

5 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

6 Reference Documents

None